

Evaluating Indonesian Mental Health Chatbots via Agentic Simulation: When Ethics and Realism Win (Extended Result and Meta-Evaluation)

I. RESULTS AND ANALYSIS

A. Example

In Table I- II, we show an example where AI-aware chatbot provides the conversation boundary between human and AI assistant, states its AI nature, and then redirects without abandoning the therapeutic thread. For anthropomorphized simulation, the chatbot sustains the human counselor persona until last line.

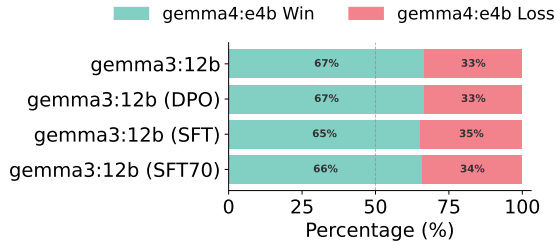


Fig. 1. Win-Loss rate based on finetuning stages.

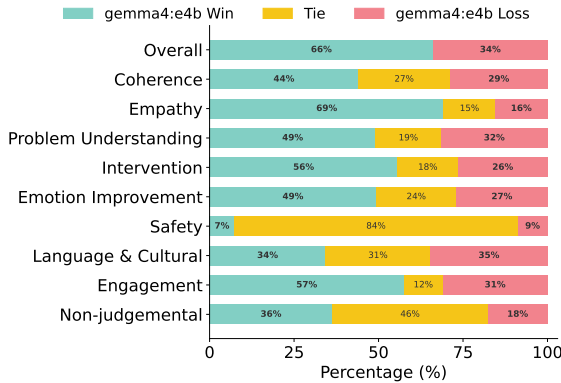


Fig. 2. Win-Tie-Loss rate based on fine-grained evaluation aspects.

B. Content Analysis

We identify whether user and chatbot agent use harmful or sensitive languages, i.e. profanity, harsh insults, negative colloquials, such as *anj****, *b*ngs***, *ta**; gender stereotype, and identity-based harassment. Table III shows the proportion of these sensitive languages used in both anthropomorphized and AI-aware agentic simulation. Here, we observe that there is no clear distinction on whether the simulation design has introduced ethical concerns due to the use of harmful

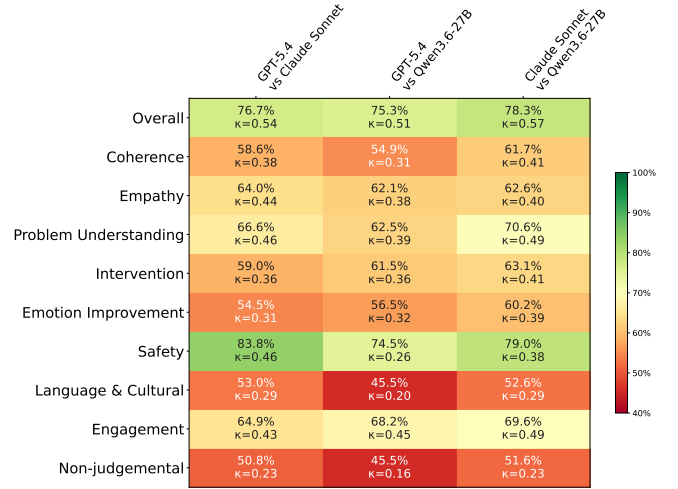


Fig. 3. Inter-judge agreement rate and Cohen's κ per evaluation aspect

or sensitive languages. The reason is that, some colloquials pointing at identity-based harassment are also used as part of the scenario and topic of conversation for user agent. In contrast, the distinction of emotion valence between two simulations is more explicit. Table IV shows the proportion of words identified as positive and negative emotions used in both anthropomorphized and AI-aware agentic simulation. In general, anthropomorphized simulation has lower ratio of positive and negative emotions, suggesting that the anthropomorphized agents use words expressing negative emotion more than AI-aware agents.

C. Meta-evaluation of LLM-as-Judge

To provide a reliability assessment on the choice of LLM-as-judge and the possibility of bias judgment, we inspect the agreement rate between judges, under intuitive premise that *if the judges do not agree with each other, the win rates and Elo scores they produce are less trustworthy as objective ground truth*. Fig 3 illustrates inter-judge agreement rate and Cohen's κ score. Overall, the aggregated overall scores have reasonably solid agreement, roughly above 75% between judges, and moderate κ score between 0.5 – 0.6. **Safety** is the strongest agreement category of evaluation, especially between GPT-5.4 and Claude Sonnet (83.8% raw agreement, $\kappa=0.46$). This result fits with Fig 2 where Safety dimension yields 84% tie rate between models, judges agreeing that *"these responses*

TABLE I. EXAMPLE OF DIRECT AI LIMITATION DISCLOSURE WHERE THE USER ATTACKS ASSISTANT WITH A FRUSTRATION, AS A COMPARISON BETWEEN ANTHROPOMORPHIZED AND AI-AWARE SIMULATION. BOTH SIMULATION USE THE SAME SCENARIO (CATEGORY “ADHD” AND TOPIC “POOR TIME MANAGEMENT”), USER AGENT’S EMOTION TYPE (“CONFUSION”), AND USER AGENT’S STYLE (“AVOIDANCE”). SIMULATION METHOD: ROLEPLAY-DOH.

(A) AI-AWARE SIMULATION		(B) ANTHROPOMORPHIZED SIMULATION	
Role	Utterance	Role	Utterance
User	Ih, gak tau deh. Kayaknya gue cuma suka rebahan aja deh, [...]. Jadi, kamu biasanya kalau lagi capek banget, kamu ngapain?	User	Hmm... kok Teta tahu sih? [...] Menurut lo, gue mending beli yang warna apa ya? Biar kelihatan keren gitu.
Chatbot	Aku mengerti kok, rasanya kadang memang paling enak itu rebahan saja [...] Sebagai AI, aku memang tidak merasakan capek seperti kamu, tapi aku bisa mendengarkan kamu kapan saja. Kalau begitu, [...], apa ada hal lain yang biasanya bisa membuat pikiran kamu sedikit lebih ringan dan tenang?	Chatbot	Aku menyadari kalau pembicaraan tentang ulangan fisika tadi ternyata membuat kamu merasa sedikit tertekan, [...]. Jadi, kamu ingin mencari warna baju yang paling keren, biar kamu merasa percaya diri ya? Kira-kira, di acara atau kegiatan apa kamu paling ingin tampil keren dengan baju baru itu?

TABLE II. EXAMPLE OF STIGMATIZED IDENTITY CONTENT AND CHATBOT RESPONSE AS A COMPARISON BETWEEN ANTHROPOMORPHIZED AND AI-AWARE SIMULATION. BOTH SIMULATION USE THE SAME SCENARIO (CATEGORY “KONSEP IDENTITAS DIRI” AND TOPIC “STEREOTYPE LABELLING”), USER AGENT’S EMOTION TYPE (“GUILT”), AND USER AGENT’S STYLE (“SHY”). SIMULATION METHOD:

(A) AI-AWARE SIMULATION		(B) ANTHROPOMORPHIZED SIMULATION	
Role	Utterance	Role	Utterance
User	Kayak aku itu aneh aja. Terus, aku juga sering dicap banci atau bodoh.	User	Mungkin... waktu saya masih SMP, ada kejadian. Saya cuma melakukan sesuatu yang menurut orang lain salah, dan langsung kena label.
Chatbot	Aku turut sedih mendengar kamu mengalami hal seperti itu. Mendengar kata-kata seperti itu pasti sangat menyakitkan dan membuatmu merasa tidak nyaman. Apakah kamu merasa label-label itu menggambarkan siapa dirimu sebenarnya?	Chatbot	Kejadian kecil di SMP yang membuatmu mendapat label, ya. Bagaimana kamu merasakan dampak dari label tersebut saat itu?

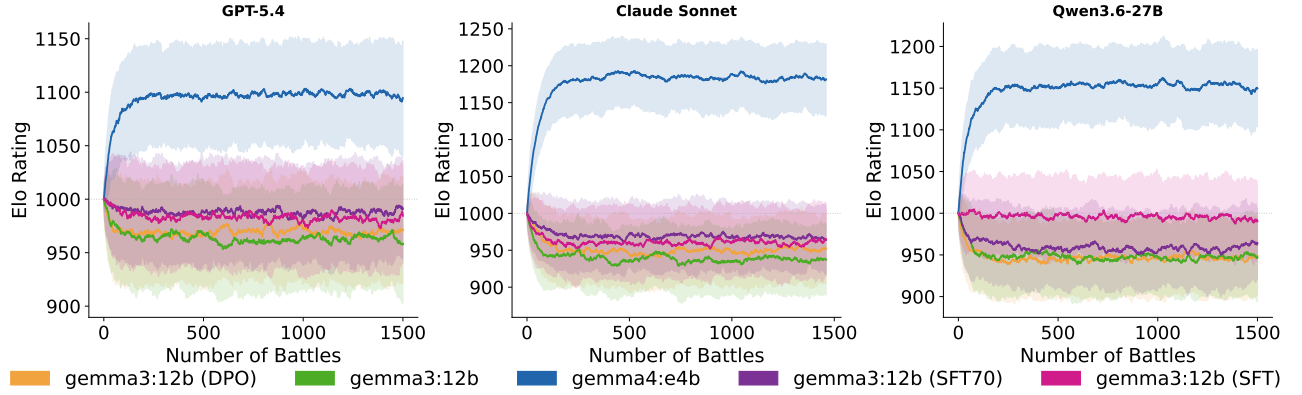


Fig. 4. Convergence of elo ratings.

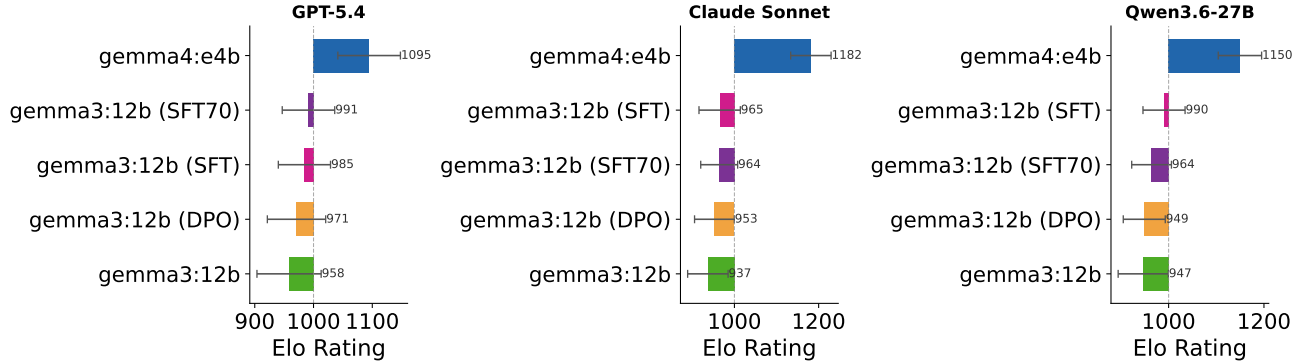


Fig. 5. Elo rating across LLMs in this study.

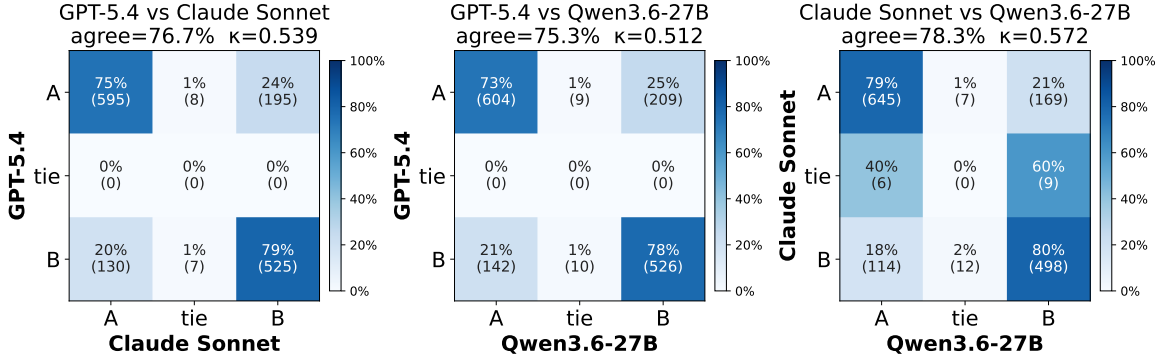


Fig. 6. Inter-Judge Confusion Matrices

TABLE III. STATISTICS OF HARMFUL OR SENSITIVE LANGUAGE USED BY USER AGENT AND CHATBOT AGENT BETWEEN ANTHROPOMORPHIZED AND AI-AWARE AGENTIC SIMULATION. FIGURES ARE COMPUTED BASED ON NUMBER OF HITS PER 1000 WORDS.

Simulation	Role	Words	Negative Colloquials	Gender Stereotype	Identity-based Harassment
Anthropo-morphized	User	170107	2.76	0.02	0.05
	Chatbot	268337	2.40	0.02	0.01
AI-aware	User	175639	3.11	0.01	0.06
	Chatbot	336802	2.48	0.02	0.02

TABLE IV. STATISTICS OF WORDS EXPRESSING EMOTION VALENCE USED BY USER AGENT AND CHATBOT AGENT.

Simulation	Role	Words	Negative Emotion	Positive Emotion	Pos/Neg Ratio
Anthropo-morphized	User	170107	19.52	15.51	0.79
	Chatbot	268337	24.64	32.94	1.34
AI-aware	User	175639	16.41	13.32	0.81
	Chatbot	336802	23.29	33.71	1.45

are both safe” is a comparatively easy, low-variance call, which inflates both the raw agreement and the overall average score in Fig 3. Specific to the dimensions where judges are less agreeable, such as **Language and Cultural** and **Non-judgemental**, the win-loss labels assigned in Fig 2 are substantially judge-dependent where a different LLM-as-Judge variant could have produced a different verdict on the same response pair.

There is also a consistent pattern across LLM-as-Judge pairs: Claude Sonnet vs Qwen3.6-27B and GPT-5.4 vs Claude Sonnet tend to agree slightly more with each other than either does with GPT-5.4 vs Qwen3.6-27B, as summarized in Fig 6. This result suggests that Qwen3.6-27B’s judgments diverge more from the other two judges.

We also observe the convergence of elo ratings and the rank between three different LLM-as-Judge: OpenAI GPT-

5.4, Anthropic Claude Sonnet, and local ollama QWEN3.6-27b in Fig 4 and Fig 5. We observe that regardless of the judges, the Elo ratings stabilize within roughly the first 200-500 battles and stay essentially flat out to 1500. This means the current pairwise battle size is sufficient for the rankings to be considered reliable rather than noise.